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B.E - (Computer) (Sem-VII)(Revised Course 2019-2020)**EXAMINATION JANUARY 2023****Data Analytics****[Time: 3 Hours]****[Max. Marks:100]**

- Instructions:** 1) Answer ANY 2 QUESTIONS from **Part A & Part B** & Any 1 from Part C.
 2) Make suitable assumptions wherever needed.
 3) Figures to the **right** indicate full marks.

PART A

- Q1 a) Explain: Profile of a Data Scientist, Also Provide set of skills they possess & Behavioral characteristics. 6
 b) Explain K-means clustering algorithm with its mathematical formulation. Also provide a relevant example. 8
 c) What is multiple linear regression? State how to use multiple linear regression with its relevant formula's 6
- Q2 a) Explain in Brief: Typical analytical architecture. 7
 b) What does random refer to in Random forest? Explain its working. 7
 c) What is Apriori principle? How many phases are in association rule Apriori algorithm. Explain. 6
- Q3 a) Explain K-medoids as a clustering algorithm. Is it better than K-means? 8
 b) Write the ID3 Algorithm. Explain its steps and advantages. 6
 c) Explain in Brief: Steps in text analysis process. 6

PART B

- Q4 a) Explain the R nuts and Bolts in Detail. 6
 b) Explain in Brief the following Types of NoSQL Databases: i) Key-value Pair Based, ii) Column-oriented Graph, iii) Graphs based, iv) Document-oriented. 8
 c) What is Data Analytics life cycle? State its different phases. 6
- Q5 a) What are the different Data types Supported in R, Explain them in Brief. 7
 b) Can R be used for Data Visualizing? Explain the different ways to do so. 8
 c) What are the Advantages of NOSQL Databases? 5

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- Q6 a) Name & Explain the following: i) Basic operations, ii) Basic statistics pertaining to R. 8
- b) What are the different database languages? State the importance of each. 5
- c) Differentiate between SQL & NoSQL. How to decide which is relevant for a project development. 7

PART C

- Q7 a) Explain in Brief: Chi Square Test by providing the formula for the same. 6
- b) State the different principal functions used to get the data into & out of R. 6
- c) Explain the SVM Classification algorithm. State its properties, functions and types 8
- Q8 a) Write a short note on T- Test. 6
- b) Write a short note on Statistical hypothesis generation and testing. 7
- c) What is re-sampling? State and explain the different re-sampling methods 7